

APPROVED

GA in MSc Cyber Security
School of Computing, Engineering and Technology

Version	2.0
Course Type	Postgraduate
Award Type	Graduate Apprenticeship in Master of Science - GA in MSc
Awards	
No Course Award Assigned	
Code	0636
SCQF Level	Level 11
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	180
ECTS Credit Points	90
Valid From	2025 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100365	computer networks	computer networks	100%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

What the Course Involves	The Eighteen-month Graduate Apprenticeship in MSc Cyber Security is an industry focused degree route, where students work in course-relevant employment for a partner employer while studying towards a fully accredited degree. The course combines workplace training with a programme of education designed to meet students' career ambitions. The core curriculum is focused on latest thinking in cyber security coupled with a "flipped-classroom" approach to work-based learning developed through ongoing collaboration between University staff, Workplace Mentors, and line managers in the workplace. Learning support materials are provided through the VLE which includes a range of equipment such as video and online learning technologies as well as online interaction. Where appropriate, modules use technologies such as computer-mediated formative assessment to bring interactivity and immediacy to the teaching and learning process. Additionally, there are opportunities for the student to attend scheduled on-campus events to allow face-to-face interaction with staff and peers, participate in group activities with peers and this contributes
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	to the overall student experience. Staff support students engaged in problem solving from an academic and/or theoretical perspective, while Workplace Mentors guide students in the application of theory to workplace scenarios. University staff undertake visits to the student's place of work in order to observe the work-based learning environment and discuss the student's progress with the Workplace Mentor. This combination of blended academic study allows the student to contextualise learning immediately within their workplace thereby developing their understanding and enhancing and their contribution to organisational effectiveness.
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Course Collaborations	
No collaborations listed	

Introduction to the Course	Graduate Apprenticeships provide work-based learning opportunities for employees and have been created by Skills Development Scotland in partnership with industry, the Scottish Funding Council (SFC) and the Further and Higher education sector. Graduate Apprenticeships combine academic knowledge with work-based learning to enable students to become more effective and productive in the workplace. They provide a new way into degree-level and Masters level study for individuals who are currently employed, and afford the opportunity to undertake degree study whilst working and earning for those individuals who may otherwise have sacrificed Secondary or Further education to enter the world of work. They enable employers to develop their workforce and support staff to build their skills to industry and professional standards. Since Graduate Apprenticeships are designed around the needs of industry, employers can have confidence that what staff are learning will directly contribute to the success of the business. As they are in continuing employment, students can directly apply academic learning to workplace situations, thus truly bringing their studies to life. The Graduate Apprenticeship in MSc Cyber Security provides students with the knowledge, understanding and skills required to become a successful cyber security professional. The course covers topics which offer students a thorough grounding in information risk management, development of a successful security program, appraisal of security controls to meet business requirements, building and maintaining of secure networks, development and execution of vulnerability assessments and penetration tests, security operations and digital forensics. Students will be able to help protect their organisations against evolving threats using state of the art approaches and technologies in cyber security. They will develop the lifelong learning professional skills, ethics and behaviours required to stay abreast of a rapidly evolving domain. Depending on the nature of the organisation, typical examples of career opportunities across a broad range of sectors, might include Information Security Auditor, Information Security Compliance Officer, IT Risk Specialist, Ethical Hacker/Penetration Tester, Security Operations Centre Analyst/Manager, Network Manager, IT Security Engineer, and Digital Forensics Analyst. However, in many organisations, there may be specific elements of the degree content involved in the role which are not reflected in the job title but which are a key component within it. The Graduate Apprenticeship allows the student to balance academic and workplace learning effectively. With the University's supported online learning platform and tools, students have flexibility to engage in their academic studies around personal and organisational commitments. Attendance on campus is minimised to reduce workplace disruption and costs. Online learning can be accessed 24/7, is varied and relevant to the topics undertaken and includes the use of multi-media interactive activities, podcasts, quizzes, webinars and forum discussions, thus providing an engaging and motivating learning experience. On-campus workshops will introduce and consolidate online learning, and support the student in preparation for academic assessment. Students are supported through the course to achieve professional excellence through a suite of modules designed to develop academic, personal and interpersonal skills. They will be encouraged to develop an enquiring, analytical, open minded and creative approach to foster independent judgement, critical self-awareness, ability to reflect, and an appreciation of the wider context of the course of study. These skills will be developed and practiced both in the context of the academic environment and in their workplace organisation.
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Educational Aims of the Course	The purpose of the Graduate Apprenticeship in MSc Cyber Security is to provide students with the required knowledge and understanding of information security principles and techniques. The development of this knowledge is reinforced with the development of professional, research, and communication skills, enabling students to become cyber security experts equipped to perform effectively in any working environment. Specifically, the aims of the course are as follows: • To provide students with specialist, advanced skills in cyber security. • To produce specialist practitioners in the design of secure information systems and networks. • To provide students with the knowledge and skills required to plan and deliver a successful risk-based security program aligned with their work-place requirements. • To provide students with the knowledge and skills required to manage computer facilities with a view to preventing, detecting and recovering from security breaches. • To make the students aware of the ethical, social, legal and economic considerations related to cyber security. • To develop the students' professional, research and communication skills. • To develop the students' commitment to CPD and reflective practice such that they constantly seek to enhance their skills in a rapidly evolving domain. • To integrate the programme with the students' experiential learning and training as part of an apprenticeship with their employers. The Graduate Apprenticeship student journey differentiates itself from the traditional course of study in that students will be employed for the entire duration of their educational journey. Their employment environment affords them a unique opportunity to take full advantage of work-based learning opportunities, appreciate the relevance of theoretical learning, and open up future career opportunities with their employer. Activities within the employing organisation are coordinated to reflect the content of the modules in which students are enrolled, and are designed to provide the practical application of the theory received through academic learning. The work-based learning within the course aims to offer a new learning experience to meet the needs of both individuals and employers. This will play a role in achieving the aspiration of the Scottish Government to see a culture of real partnership between employers and education and increasing skill levels in the workforce. This is entirely in keeping with the commitments of the University to 'extend the reach and relevance of learners' opportunities to gain employment and thrive in their professional career'; to 'develop the range and diversity of the University's provision to support individuals' professional development'; to 'ensure a high quality student experience'; to 'adopt a student-centred approach to curriculum design and delivery'; to 'work with organisations to enhance skills'; and to 'expand the range and depth of partnerships with employers and the professions'. The approach taken by the University in respect of flexible admissions requirements, a commitment to helping partner employers ensure equality of opportunity, dedicated study skills support for learners from the point of induction, innovative curriculum design and delivery and work-related/centric assessment is supportive of the University's equality and diversity ambitions.
Awarding Institution	Robert Gordon University
Course Duration	Work-Based Learning Full-Time - 18 months with a September intake. 18 months with a February intake (availability subject to funding as per Course Specific Academic Regulations).
	The course integrates the expertise of staff gained from research, consultancy and scholarly activity into the programme materials. The School of Computing, Engineering and Technology boasts state-of-the-art facilities including a dedicated cyber security lab (36 seats) and a networking lab (36 seats) linked via fibre links and equipped with racks of Cisco equipment. The School also enjoys excellent external linkages with industry, including its Industry Advisory Board, and the hosting of the Aberdeen Cyber Security Meetup. Moreover, the School is an academic partner of the EC-Council and a recognised test centre for the Certified Ethical Hacking certification, and is also part of the Cisco academy.

Distinctive Features of the Course	The mode of delivery is a flexible student-centred approach to studying allowing for business and personal life commitments to be considered and planned for in advance of key deadlines. This course shares the institutional Graduate Apprenticeship characteristics; therefore course design: • optimises candidate time within the workplace; • is flexible, enabling students to participate in the course and student experience whilst not bound to a physical campus; • includes meaningful and substantial employer engagement from the outset; • uses innovative teaching, learning and assessment methods; • supports the development of reflective practitioners, equipped to excel in their field and add value to their employers; • ensures the professional practice of learners is informed and enhanced by expert technical and theoretical knowledge; • combines work-based learning, the latest thinking within the field, and work-related/ centric assessments to produce graduates equipped with valuable skills and knowledge. The Graduate Apprenticeship model of achieving a degree through work-based learning has been designed such that: • In contrast to the traditional student journey, Graduate Apprenticeship students are employed for the entire duration of the course, and the course is designed to integrate course learning outcomes with suitable workplace projects in collaboration and partnership with their employer. The development of transferable skills is also supported throughout the course via course content, assessment, and work-based learning. • The course utilises the technology investment the University has made in lecture capture and Virtual Learning Environment (VLE) delivery providing the student with a cutting edge experience. Employers are closely involved with the specification and delivery of work-based learning activities to support the learning and development of students. Workplace Mentors appointed by the employer, and supported by the University, work with students to create individual learning plans which outline the shadowing, observation, training, work activities and project opportunities which students undertake in the workplace. These learning plans are reviewed by the University to ensure the required employer contextualisation is appropriate. The course adopts a practical approach to developing students' professional skills; for example employing group discussion activities within course workshops to foster leadership, listening, and communication skills, and requiring students to deliver presentation and reports as a means of further refining their professional skillset. Within the work-based learning environment, students will continue their professional development as they enhance planning and negotiation skills via development of their learning plan in conjunction with their Workplace Mentor and engage in practical activities designed to further their professional skillset. The knowledge and insights gained across the University and work-based learning environment, coupled with the active role students are expected to take in planning and executing their own learning, will act to support the students' overall professional enrichment.
Placement/Professional Experience	Students enrolled on the course are employed on a full or part-time basis (minimum 21 hours per week) by a collaborating partner organisation while studying for the degree. Employers may provide a series of internal placements within their organisations as part of the student's learning plan where it would be beneficial for the student to spend time within a department other than their own to gain specialist skills and knowledge and where this can be supported by their employer.
Opportunities for Further Study	A Masters degree can provide access to doctoral programmes within this institution or other UK higher education institutions.

External and Internal Indicators of Quality and Standards	In October 2016 the QAA (Quality Assurance Agency for Higher Education) issued an Interim Statement on Degree Apprenticeships confirming the final UK-wide statement will be in keeping with the existing suite of characteristics statements within Part A of the UK Quality Code for Higher Education. The QAA further state the Quality Code is a key reference point for setting the academic standards and assuring the quality of apprenticeships involving a higher education qualification. Subsequently, in May 2017 the QAA published its paper 'Quality Assuring Higher Education in Apprenticeships: Current Approaches', a second edition of which was published in July 2018. This document highlights considerations for higher education providers in relation to standards and quality and extrapolates relevant guidance from the UK Quality Code. Whilst the document does not constitute formal QAA guidance, its contents have informed the University's approach to the design and delivery of all Graduate Apprenticeships. QAA published a Computing Subject Benchmark Statement in 2019 outlining the standards expected for the award of Master's degrees in computing which, together with the Skills Development Scotland Cyber Security subject framework, provided the basis for the development of the Graduate Apprenticeship in MSc Cyber Security. The QAA benchmarking statement for Computing represents the national threshold standards for achievement in this area and articulate the attributes and capabilities, which those possessing qualifications from these areas should demonstrate. As such, QAA benchmarks underpin course articulation, delivery and development through providing guidance for learning outcomes across the breadth of the course and within specific cognate areas such as technology and management. SCQF statements are also used during the design of modules, courses and programmes and the relevant SCQF level is included within each Module Descriptor. The University appoints External Examiners to assure standards of assessment. Examiners are appointed from both academic and practice backgrounds to cover key subject areas within Computing. The School of Computing, Engineering and Technology has strong links with other academic institutions, both within the UK and Europe, which supports the maintenance of standards at national and international levels. In November 2018, the School had its Chartered IT Professional accreditation renewed by the BCS (British Computer Society) for its undergraduate and postgraduate programmes. The quality assurance procedure includes a regular cycle of accreditation visits by a panel of subject experts appointed by the BCS who evaluate the courses according to strict quality control standards. In April 2021, the School's MSc Cyber Security course received full certification by the National Cyber Security Centre (NCSC), making it only one in three masters courses in Scotland to have achieved this accolade. This course benefits from the accreditation as MSc Cyber Security teaching staff will also teach on the Graduate Apprenticeship. A number of internal processes and procedures are used by the school to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Postgraduate Programme Management Team, Student Experience questionnaires, Industrial Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure

	Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Employment: All applicants must be employed in a full or part-time position (minimum of 21 hours per week), relevant to their course of study and have the right to live and work in Scotland. An applicant's employer must be committed to and involved in the provision of a suitable workplace environment, coupled with supportive workplace learning guidance and mentoring to create a suitable workplace environment in which the student will be able to obtain the necessary experience and learning outcomes. Employers must be willing to formally partner with the University via a Collaboration Agreement which sets forth the obligations of each organisation and an individual-level Learning Agreement which details the obligations of the Workplace Mentor; Line Manager if they are not also the Workplace Mentor; Student; and University. Subject specific qualifications: Applicants normally hold a 2:2 undergraduate honours degree, or its equivalent, in a subject area with substantial Computing content. English language requirements: All applicants whose first language is not English are required to have an Academic IELTS (International English Language Testing System) minimum score of 6.0. Appropriate equivalents will be considered.
Alternative Qualifications for Entry	Applicants with a degree in an alternative subject area or lower degree classification and/or with relevant work experience will be considered on a case by case basis.
Interview Procedures	Generally, an interview is not required for admission to the course. However, the Course Team may request to interview applicants with alternative qualifications for entry.
Course-Specific Regulations	Due to the sequential delivery of the modules and the scheduling of the Assessment Board, students will be allowed one re-assessment opportunity, for their taught modules, prior to the August Assessment Board. The main intake for the course is in September. A February intake may exceptionally be available depending on annual funding availability. Please contact the School for further information on intakes.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU.

	All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
	The introduction of the Graduate Apprenticeship model is a natural development of the University's philosophy to combine theory, practice and experience in developing the skills, knowledge, confidence and professional practice of its students. The principles of work-based learning are core to the philosophy of each Graduate Apprenticeship course at Robert Gordon University. It is fundamental to the Graduate Apprenticeship courses that students have access to relevant and up-to-date theory as they approach practical workplace situations and problems. Encountering theory and practice simultaneously will

Course-Specific Learning Experience	enable the student to critically analyse each in conjunction with the other. This will embed a rounded knowledge base and develop the ability to question situations while seeking a range of opinions in forming solutions to challenges. Working in conjunction with learning will also enable students to develop the soft skills necessary in an ever-changing workplace. This means that the student will develop not only the knowledge and skills required for the technical aspects of their career but also the ability to operate in the workplace and communicate effectively with internal and external parties. Most of the practical aspects of the course take place in the student's workplace and will therefore vary according to the nature and resources of the student's employing organisation. In addition, the resources of the School of Computing are available to the student and support of the teaching staff attached to the course.
	The institutional teaching and learning strategy for Graduate Apprenticeships is characterised by a commitment to ensuring a genuine interaction between theoretical and work-based learning. This is supported by a pedagogical approach which embraces and makes use of experiential learning where experiences framed by reflection lead to learning (Schon, 1983) and andragogy where adult learners understand the need to learn something and self-direct their learning process via a negotiated learning plan. Specifically students are encouraged to consider learning objectives; identify learning opportunities associated with these and reflect upon these experiences to consolidate and apply learning. With regard to work-based learning activities, employers are guided in the identification of suitable activities and projects, and Workplace Mentors are supported in aiding students to reflect on and capture learning from the workplace and encourage the unification of knowledge and experience from the work and university spheres. The Learning Process: Teaching Strategy and Methods The course curriculum is designed to deliver the skills and knowledge identified by Skills Development Scotland's framework for Cyber Security. It has been developed as a coherent whole with an emphasis on the integration of theory, practical skills and applications. It enables the student to acquire and develop the subject knowledge and understanding necessary for successful professional employment. Close consultation with the student's employer will ensure that practical situations and projects encountered within the workplace correlate with the theoretical knowledge developed throughout the course. Opportunities for critical reflection and analysis of apparent or real disparities or conflicts between theory and practice will inform the student of the practicalities of real-world situations and form a rich basis for developing their professional practice. As the student progresses through the course they are encouraged to expand their understanding and critical appreciation of all aspects relating to the study of Cyber Security. A central feature of all Robert Gordon University Graduate Apprenticeship courses is that they promote a unified, integrated view of the subject area. Courses are designed to: • Present a coherent underpinning of the theory and practice appropriate to the aims of the course which is further developed and applied within the work-based learning environment. This enables graduates of the course to adapt to future developments in the field. • Reflect changes within the field and ensure that coverage is given to a selection of emerging topics so that students are aware of likely future developments in the subject together with their potential impact. • Integrate the delivery of teaching across thematic boundaries to highlight fundamental aspects of interconnectivity and interdisciplinarity within different areas of the syllabus. • Integrate modes of assessment which complement the methodology of work-based learning and support the development of workplace communication skills. • Facilitate students to exhibit competence and achievement across a number of different themes (integrated assessment). • Encourage students to combine elements of their learning from different parts of the course and to show their accumulated and integrated knowledge and understanding of a topic or subject area (synoptic assessment). The professionally-orientated nature of the course requires students to obtain general and specific skills, this being addressed by a combination of teaching and learning methods. These methods are built around mentored workplace learning directed, supported and extended by University guided, student-centred learning including virtual lectures, tutorials, discussion

Teaching and Learning Strategy

forums, workshops and private study.

Teaching and Learning Approaches

The main goal of the teaching and learning strategy for the University's Graduate Apprenticeships is to enable students to become critical, reflective, independent learners who are active participants in the learning process and who develop the motivation and autonomy needed to take control and regulate their own educational experiences and develop the discipline of continuous professional development and life-long learning. This means that, throughout their course of study, students will begin to take more responsibility for their own acquisition of necessary skills and knowledge so that, by the end of their degree, they will be in a position to evaluate and assess their own effectiveness.

This process is driven by various objectives such as the need for constructive alignment of syllabus and learning approaches and makes use of a wide-range of appropriate delivery styles. In addition to mentored workplace learning these include conventional instructional teaching methodologies, social learning activities such as group discussion, as well as constructivist, investigative approaches such as problem-based learning and enquiry-based learning.

The delivery of material centres on using the university's VLE, CampusMoodle, virtual campus platform. This centres on the creation and sustenance of active community groups, and delivery of topics is structured in such a way as to facilitate the maintenance of student interaction. Papers, supported by recommended reading lists and private study guides, are augmented by a discussion forum and the initiation of discussion threads. Live chat rooms are used to encourage debate and the exchange of ideas between students, on the one hand, and with tutors on the other. Video conferencing is used, as required, during specific workshops and presentation/discussion of project work. The Library provides support for all students and is progressively expanding its provision of electronic texts such as e-journals and e-books.

Modes of Delivery

Work-Based Learning

The workplace forms the central learning environment for the student. It is important, however, to appreciate that whilst much learning takes place naturally in a workplace that this typically happens in an unplanned and unstructured way, with limited opportunity for reflection or consolidation of learning. For the purposes of this course there needs to be a structure in place to guide the student and ensure that learning outcomes are met. This structure is governed by a subject framework established by Skills Development Scotland and developed into a curriculum by the University who work closely with the student's employer to ensure that learning outcomes are achieved through correlated theory and practice. This is expressed for the student in the form of a learning plan which guides them through the course and helps to ensure that workplace learning is supported, resourced, and planned with space and time for critical reflection and consolidation of learning.

Workplace Mentoring

Key to the delivery and support of work-based learning is the Workplace Mentor. Partner employers are required to assign a Workplace Mentor whose role is to oversee the work-based learning undertaken by the student. The University provides employers with guidance as to how appropriate Workplace Mentors should be selected, and provide Workplace Mentors with training and support via induction and ongoing liaison designed to help mentors understand the remit of their role and the activities associated with this. Workplace Mentors have an overview of the course and its aims, develop the student's learning plan, assign projects and in-house placements and provide feedback on work-based activities. Workplace Mentors are granted 'Affiliate Student' status enabling them to access all course learning materials via the University's VLE. In addition to mentoring support for students, supplementary support may be provided by the employer, for example when subject matter experts are brought in to provide specific business or technical expertise or through access to other departments within the organisation. This enables work-based learning to reflect the range of the curriculum as the student progresses in their studies, therefore providing adequate support for the student's workplace learning on a project, placement or task basis.

Employers ' Responsibilities

Partner employers commit to supporting students via the signing of a Collaboration Agreement which sets forth the obligations of the employer and the University. In supporting students, employers are obliged to provide a work-based learning environment which adequately represents and reflects the students' course to ensure appropriate functional expertise, supporting infrastructure and professional and technical knowledge within the organisation. Central to this is the provision of a supportive and educative work-based learning environment which affords opportunities to learn, gain new experiences, acquire and build competencies, and reflect on and apply learning. Additionally, employers are obliged to provide protected time to allow students to attend on-campus study days and to undertake study-related activities such as participation in online discussion forums, and access to the requisite IT hardware and software to facilitate this.

Student Role in Learning

Students are expected to be active participants in the learning process. This means that there is a requirement for students to extend their learning from formally delivered materials through wider reading, research and self-directed enquiry. As students develop over the duration of the course, they are expected to become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their own personal objectives for learning. Students are encouraged to reflect regularly upon their learning and to establish a view of their own progress and development, which they triangulate with feedback from their Workplace Mentor and course delivery team. Students are expected to utilise the full range of facilities and resources available to them, for example the services and resources provided by the library and the support available via the Graduate Apprenticeship Success Coach.

Lecture s

Lectures are for the purpose of dissemination of information and the demonstration of techniques. These are usually be available via the VLE but may also be delivered face to face at the University as part of an on-campus event.

Tutorials

Tutorials may be arranged remotely (e.g. online via the VLE) or in person. Tutorials are an opportunity to raise questions, discuss ideas and consequences and exchange experiences and lessons learned. This format is used for a wide range of activities, each suited to the particular subject. It may consist of the staff supporting students engaged in problem-solving or may consist of a series of group exercises where each group is encouraged to allocate responsibilities, allocate tasks, etc. Generally this type of teaching is used to support topic content, clarify the material and experiment with the techniques and skills required.

On-Campus Events

Beyond the initial course induction, Graduate Apprenticeship students are expected to attend 1 to 2 on-campus day per module of study. The format of these days vary accordingly to reflect course content and student needs, but generally these may be group sessions where students can discuss the relationship between theory and practice through critical analysis and reflection. On-campus events also give students the opportunity to develop their professional practice through ideas exchange with peers and academic staff and through other networking. On-campus events are scheduled within a course calendar which is publicised to employers and students in advance of course commencement.

Individual Project

The student will undertake a Master's Project which is an independent piece of research relating to a workplace topic relevant to the subject of cyber security. The student will agree a specification for this extended piece of work with their University tutor and their Workplace Mentor to fulfil the requirements of the Master's Project. The academic work will be completed

	under the guidance and supervision of the student's assigned University tutor. The findings from the project should shape change in the student's place of employment enhancing the project/team/departmental/business operation.
Assessment	Assessment A variety of assessment methods are used in this course, as deemed appropriate to the learning outcomes and subject material being assessed and the context in which the material is delivered. Coursework Coursework will be used as a form of summative assessment, and within the context of Graduate Apprenticeship courses this may be work-related, for example based around a project being undertaken in the workplace. Coursework will take the form of a report because it affords the greatest flexibility to academics to design authentic and innovative assessments that can be based on real or fictitious case study scenarios, practical tasks and reflective statements. Computer Aided Assessment Computer-assisted assessment packages integrated into the VLE may be used for formative assessment, to enhance student engagement and increase the speed of feedback. Individual Project Projects are the last piece of assessment which allows students to tackle more complex problems that often require the integration of knowledge and understanding, and intellectual skills from a variety of subject areas. Students will undertake a masters Individual Project which is an independent piece of research relating to a workplace topic relevant to the subject of Cyber Security. The findings from the project should shape change in the student's place of employment enhancing the project/team/departmental/business operation. The outcomes of the project will be captured in a report and will be defended in an oral presentation/viva. The Assessment of Student Work The University has developed a Grading Scheme for assessment purposes, details of which may be accessed at: http://www.rgu.ac.uk/academicregulations . It is important to note that all grades indicated to students during the course of the academic session are provisional until confirmed by an Assessment Board. The Board consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each module undertaken by them and any eligibility for an award. This is subsequently confirmed to each student in a published transcript, and may be accessed electronically on the Student Portal. Any potential barriers to students with disabilities or specific needs are supported to ensure that such students are not disadvantaged during an assessment. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made and the needs are assessed by the Inclusion Centre and appropriate teaching staff. In the case of examinations, students must register by a specified deadline date to have specific examination requirements considered.
	The University provides a number of means of supporting teaching and learning:

University Support for Teaching, Learning and Assessment	• student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student and mentor induction sessions; • an ongoing scheme of personal/pastoral support for students, including online support, such as Silvercloud, provided by the Inclusion Centre; • an extensive programme of student study skills delivered through the Study Skills and Access Unit and the Library; • an extensive library of learning resources; • close collaboration with the student's employer and with industry and professional, statutory and regulatory bodies; • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, www.enhancementthemes.ac.uk; • support for staff from the Department for the Enhancement of Learning, Teaching and Access (DELTA), including credit rated teaching, learning and assessment provision; • RGU Learning and Teaching Framework which forms a template to produce adaptable and skilled graduates equipped for the opportunities and challenges of their future professional careers. • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity; • support from the Employability Team, including online advice and information via the RGU e-Hub; • CampusMoodle, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • on-campus days in the state-of-the-art, purpose-built facilities and buildings in addition to online engagement. In addition to this, students can access study support material from the Department for Academic Quality and Learning Enhancement (AQLE). This is the central department that supports the Schools in implementing the University's strategic goals in teaching and learning, extending access and the student experience. Further information can be found at: www.rgu.ac.uk/delta . Frequent informal feedback is provided to students through regular contact with academic staff and Workplace Mentors. Feedback is provided on all assessed work normally within 20 working days of submission of the work. Provisional grades are made available electronically with final grades available electronically via the Student Portal following ratification by the Assessment Board.

University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	Annual Course Appraisals are reviewed and approved by the Course Management Team and the School Academic Board which meet at least once per semester. The Course Management Team considers teaching and learning issues arising from feedback generated by External Examiner Annual Reports, staff/student engagement/partnership liaison, and Student Voice Questionnaires.
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience.

	Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	The course cohort will be offered the opportunity to nominate a Student Representative. Student Reps work alongside the Student School Officer, who is a member of the School Academic Board where they can contribute to discussions. Students can raise issues with the Course Leader who will discuss concerns and recommend or action solutions as appropriate.

Awards

Stage 1

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Graduate Apprenticeship in Postgraduate Certificate - GA in PgCert	Cyber Security	Aberdeen, United Kingdom	Exit Award		60
Graduate Apprenticeship in Postgraduate Diploma - GA in PgDip	Cyber Security	Aberdeen, United Kingdom	Exit Award		120
Graduate Apprenticeship in Master of Science - GA in MSc	Cyber Security	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Demonstrate extensive, detailed and critical knowledge and understanding of the fundamental concepts and principles of information security.	SC1	SCQF 11
LLO2	Demonstrate a critical awareness of the organisation's business and regulatory environment and its risk appetite, including an understanding of how to align the information security governance to the organisation's objectives.	SC1	SCQF 11
LLO3	Demonstrate a critical knowledge and understanding of the principles and techniques of risk analysis and management in order to successfully plan and deliver a risk-based approach to protecting the organisation's assets.	SC1	SCQF 11
LLO4	Demonstrate a critical knowledge and understanding of the variety of key security controls and services available, including an understanding of how to appraise and select the most appropriate controls to meet given requirements.	SC1	SCQF 11
LLO5	Work with a critical awareness of the role of human factors in delivering a successful security program that involves all stakeholders.	SC1	SCQF 11
LLO6	Work with a critical knowledge and understanding of the key techniques in designing and maintaining a secure network in a variety of environments (LAN, Wireless LAN, IoT/OT), including the latest trends in network security such as the use of Machine Learning in intrusion detection.	SC1	SCQF 11
LLO7	Demonstrate a critical knowledge and understanding of the methods used in testing IT infrastructure and applications for security vulnerabilities.	SC1	SCQF 11
LLO8	Demonstrate a critical knowledge and understanding of the principles of successfully operating a Security Operation Centre and the methods used in investigating and responding to a security incident.	SC1	SCQF 11
LLO9	In conducting a risk assessment that considers the organisation's business and legal environment and the evolving threatscape.	SC2	SCQF 11
LLO10	In planning and delivering a security program following a recognised standard of Information System Management System.	SC2	SCQF 11
LLO11	In conducting a vulnerability analysis and a security test to systems, applications and networks.	SC2	SCQF 11
LLO12	In investigating a security incident and conduct its related forensics work.	SC2	SCQF 11
LLO13	In designing, implementing and managing secure systems and networks.	SC2	SCQF 11
LLO14	In applying research methods to design an experiment and evaluate its results.	SC2	SCQF 11
LLO15	In producing a clear and coherent thesis which documents the process and end result of a major project and deliver a coherent thesis defence before a panel.	SC2	SCQF 11
LLO16	Problem-solve and critically analyse facts and requirements to determine the cause of a problem and identifying and selecting appropriate solutions.	SC3	SCQF 11
LLO17	Critically evaluate and test the extent to which a security solution meets the business requirements.	SC3	SCQF 11
LLO18	Innovate, create and be enterprising, including the ability to propose own Masters project, develop and communicate ideas, and deliver successful outcomes.	SC3	SCQF 11
LLO19	Network, demonstrating an awareness of the interpersonal skills of effective listening, negotiating, persuasion and presentation.	SC3	SCQF 11
LLO20	Foster reflective thinking and critical judgement to logically analyse issues, evaluate different options and implement decisions.	SC3	SCQF 11
LLO21	Recognise the professional, economic, social, environmental and ethical issues involved in cyber security and be guided by the appropriate professional, ethical and legal practices.	SC3	SCQF 11

LLO22	Develop numeracy skills, including the use of quantitative risk analysis, and statistical techniques to design and evaluate a research experiment.	SC4	SCQF 11
LLO23	Work collaboratively with stakeholders to develop a successful security program, including security awareness and education.	SC4	SCQF 11
LLO24	Develop skills in communicating a variety of information security documents and presentations, including policies, metrics, and business cases, to a range of audiences.	SC4	SCQF 11
LLO25	Make effective use of available IT tools and packages to undertake courseworks and to produce high quality technical documentation including the Masters project thesis and presentation.	SC4	SCQF 11
LLO26	Take responsibility for own work and learning.	SC5	SCQF 11
LLO27	Exercise substantial autonomy and initiative in professional activities.	SC5	SCQF 11
LLO28	Demonstrate leadership and initiative and make an identifiable contribution to change and development.	SC5	SCQF 11
LLO29	Manage complex ethical and professional issues and make informed judgements.	SC5	SCQF 11
LLO30	Plan, manage and communicate the outcomes of a significant cyber security project.	SC5	SCQF 11

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7
		SCQF 8 - SCQF 8
		SCQF 9 - SCQF 9
		SCQF 10 - SCQF 10
		SCQF 11 - SCQF 11
		SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time Work Based Learning	

Stage 1 / Semester 1**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CMM401	Information Security Fundamentals		Hatem Ahriz (cmsha)	Yes	1.0	30	Level 11
Full Time Work Based Learning	CMM402	Network Security	Undertaken over two semesters.	Christopher McDermott (cdm7692)	Yes	1.0	30	Level 11

Stage 1 / Semester 2**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CMM403	Security Testing		Harsha Kalutarage (hk9491)	Yes	1.0	30	Level 11

Stage 1 / Semester 3**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CMM404	Secure Operations and Forensics		Ian Harris (ish7850)	Yes	1.0	30	Level 11

Stage 1 / Semester 4**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CEM100	MSc Project	Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time Work Based Learning	CMM401	Information Security Fundamentals		Hatem Ahriz (cmsha)	Yes	1.0	30	Level 11
1	Semester 1	Core	Full Time Work Based Learning	CMM402	Network Security	Undertaken over two semesters.	Christopher McDermott (cdm7692)	Yes	1.0	30	Level 11
1	Semester 2	Core	Full Time Work Based Learning	CMM403	Security Testing		Harsha Kalutarage (hk9491)	Yes	1.0	30	Level 11
1	Semester 3	Core	Full Time Work Based Learning	CMM404	Secure Operations and Forensics		Ian Harris (ish7850)	Yes	1.0	30	Level 11
1	Semester 4	Core	Full Time Work Based Learning	CEM100	MSc Project	Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
All Applicable Locations	Work-Based	Full Time	

Sources of Further Information

#	URL and Description
1	http://www.apprenticeships.scot Further information about Degree Apprenticeships in Scotland can be found at the Skills Development Scotland Apprenticeships website: www.apprenticeships.scot Robert Gordon University Graduate Apprenticeship information made available to employers and prospective students: www.rgu.ac.uk/ga Further information about the postgraduate programme in the School of Computing: www.rgu.ac.uk/pg-computing Library resources can be accessed via the Library web portal: https://library.rgu.ac.uk/home Advice for students about study and welfare issues is found at the Student Advice and Support portal:

<https://rgu.ac.uk/life-at-rgu/support-advice-services> Links to the Academic Regulations governing the course: www.rgu.ac.uk/academicregulations A number of external sources were consulted in the process of curriculum development, including: Graduate Apprenticeship Cyber Security subject framework available at: www.apprenticeships.scot QAA Computing Subject Benchmark Statement available at: www.qaa.ac.uk/quality-code/subject-benchmark-statements Professional Bodies relevant to this degree include: British Computer Society: www.bcs.org